

Constraint / random lite

Constrained random verification picks legal values automatically instead of hand-writing every vector

Rand, constraints, randomize

- A rand field tells the tool this value may change on each randomize call
- A constraint block filters the domain
- Randomize returns success when at least one legal combination exists
- The legal set here is small and discrete
- Fix a seed before you roll so two runs with the same seed produce the same sample sequence

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Constraint / random lite

Dice-level CRV literacy: rand fields, a simple constraint, randomize() success/fail, and a seed. Not a full solver. Starter: **data inside {[0:15]}** with one legal roll.

Starter example: data inside {[0:15]}, seed 42, one legal roll already in the histogram.

Load starter example

Challenges 0 / 22 cleared

Quiz: rand: A rand field is...

☐ a variable the randomize() solver may assign

☐ always a wire in the DUT

☐ only legal in always_comb

☐ a synthesis keep attribute

Show hint

Check

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Idle

1

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Core ideas

rand

Fields the solver may assign.

constraint

Legal region over those fields.

randomize()

1 = ok sample; 0 = unsat / fail.

seed

Same seed → same dice stream.

Browser lab starter

Real SV TB track practice

- In the real track, open this module's examples prompts
- Restate CRV lite in one sentence, rand fields, constraints, randomize, seed
- On paper, write a tiny class with one rand byte and constraint data inside zero through
- Predict randomize success or failure before you check
- Optional: run randomize in a local simulator if you have one

Pitfalls to watch

- Do not assume randomize always succeeds, over-constraining is a real bug source
- Do not forget the seed when you want reproducibility
- Do not treat inside as the only constraint form, modulo, inequalities
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser
- On paper, write one solvable constraint and one impossible pair
- When you are ready